

Artificial Intelligence

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Final Review Assignment

Question 1

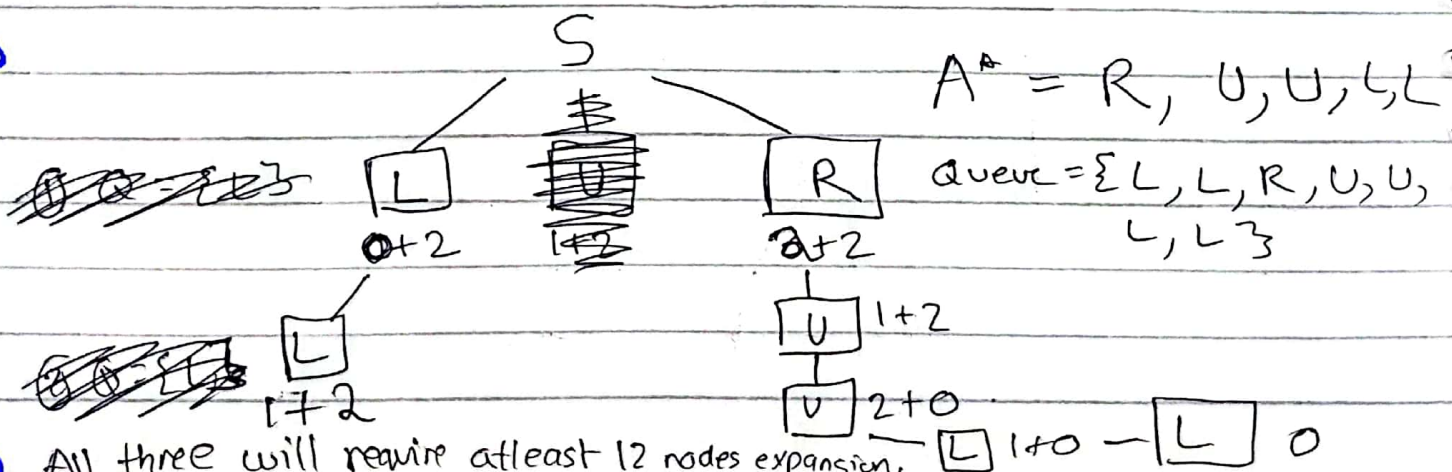
a) The state of vehicle can be represented as:-

- Current position of the vehicle
- State of the grid including position of a
- State of blocked roads might be included.
- ~~Heuristic estimate to goal~~
- ~~Path taken~~

b) The two different heuristic functions can be:-

- Manhattan distance $|x_1 - x_2| + |y_1 - y_2|$
- Euclidean distance $\sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2}$

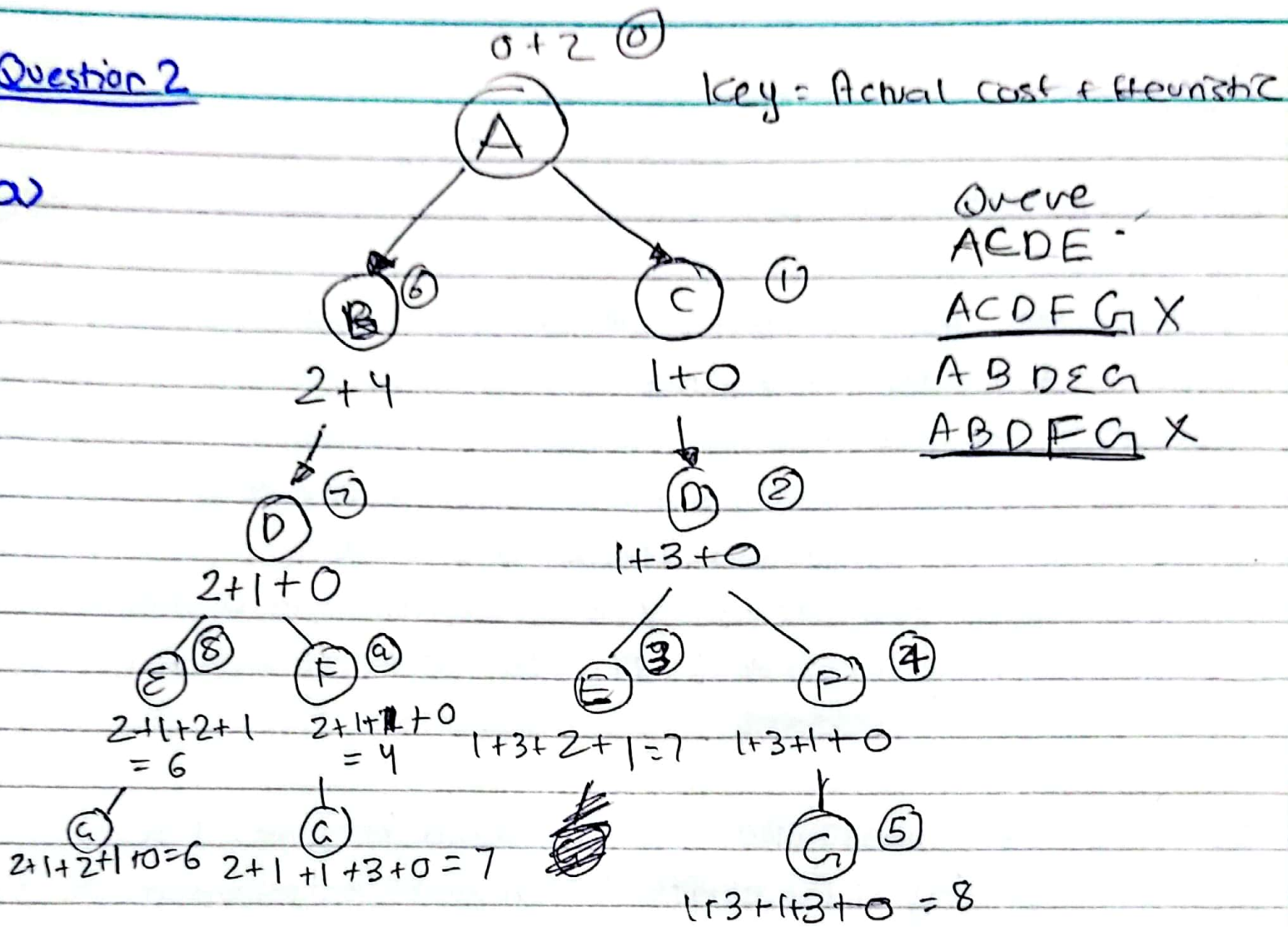
c)



d) All three will require atleast 12 nodes expansion, $\square 1+0 - \square 0$

Question 2

a)



$A^* = A \rightarrow B \rightarrow C \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow B \rightarrow D \rightarrow E \rightarrow F \rightarrow G \rightarrow E \rightarrow G$

Path $A^* = A, B, D, E, G$

c) Frontier Queue = A, C, D, E

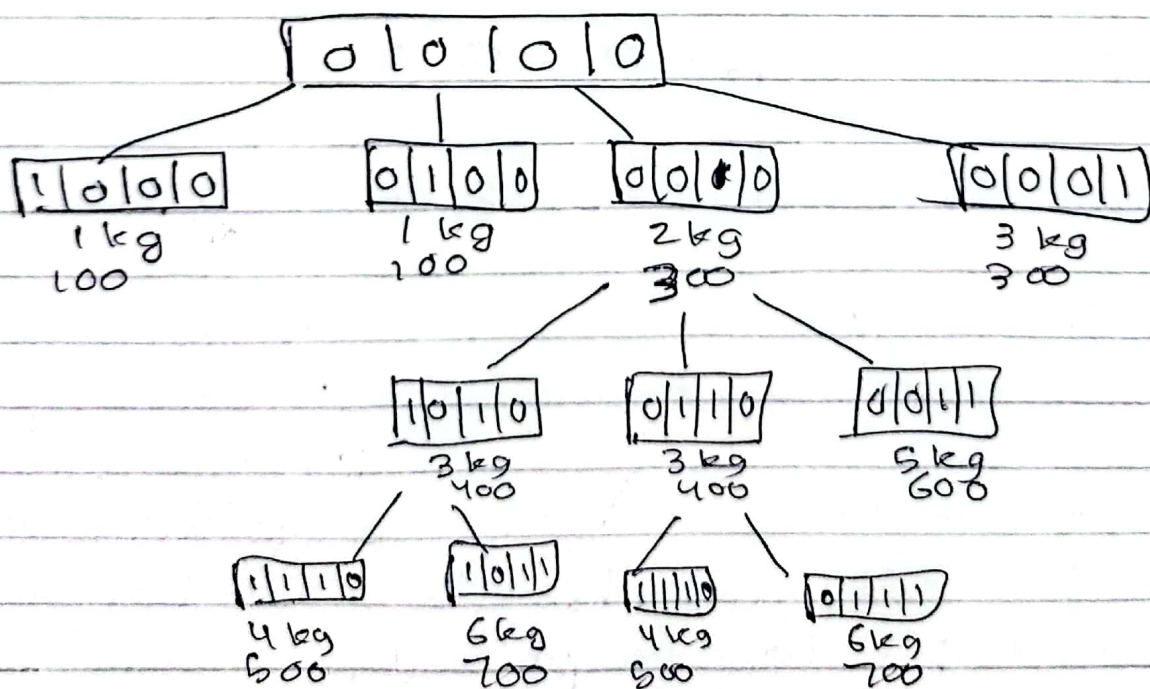
d) Path = A → C → D → E → G, the optimality will be affected now it finds a less optimal path.

e) A → C → D → F → G. The completeness will not be affected but the solution found would be less optimal.

Question 3

- a) Size of search space = 2^{100}
 $\nearrow$ Each item can be 0 or 1
 $\nearrow$ 100 items
 $\rightarrow 2^{100}$ possible states
- b) Since there are 100 items, each state can generate successors by changing one of the items from 0 to 1, provided the item is not already included in the knapsack. Therefore, the maximum number of successors a state can have is equal to the number of 0s in the array, which is in the worst case (where none of the items are initially included) is n . So 100 is maximum number of successors.
- c) A suitable evaluation function should be one that measures the quality of a state in terms of total value of items while also ensure the total weight do not exceed the max capacity.

d)



Items selected A, B, C; weight = 4 kg value = 500

Question 4

a) Training Data if $y \geq 0$ then $y = 1$ otherwise $y = -1$

Positive $n_1 = 1, n_2 = 2 \quad y = 1$

$n_1 = 2, n_2 = 3 \quad y = 1$

Negative $n_1 = 2, n_2 = 1 \quad y = -1$

$n_1 = 3, n_2 = 2 \quad y = -1$

Iteration ①

b) $w_0 + w_1 \cdot x_1 + w_2 \cdot n_2$

① $0 \times 1 + 0 \times 1 + 0 \times 2 = 0 = 1$ ~~Incorrect~~ Correct

~~Update weights $[0 \ 0 \ 0] \rightarrow [1 \ 1 \ 2]$~~

~~New weights $= [1 \ 1 \ 2]$~~

② $0 \times 1 + 0 \times 2 + 0 \times 3 = 0 = 1$ Correct

③ $0 \times 1 + 0 \times 2 + 0 \times 1 = 0 = 1$ Incorrect

Update weights $[0 \ 0 \ 0] - [1 \ 2 \ 1] = [-1 \ -2 \ -1]$

④ $-1 \times 1 + -2 \times 3 + -1 \times 2 = -9 = -1$ Correct

Iteration ②

① $-1 \times 1 + -2 \times 1 + -1 \times 2 = -5 = -1$ Incorrect

Update weights $[-1 \ -2 \ -1] - [1 \ 1 \ 2] = [-2 \ -3 \ -3]$

② $-2 \times 1 + -3 \times 2 + -3 \times 3 = -17 = -1$ Incorrect

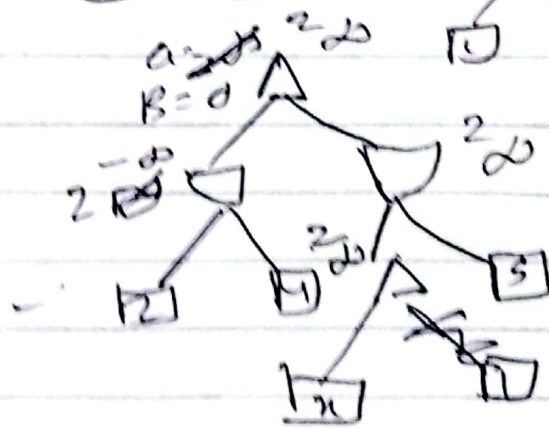
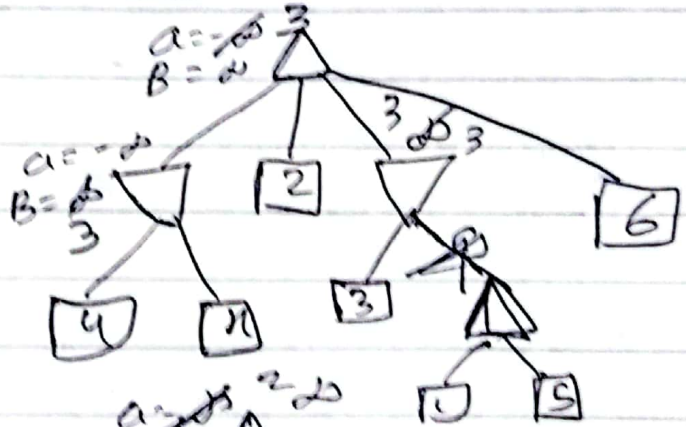
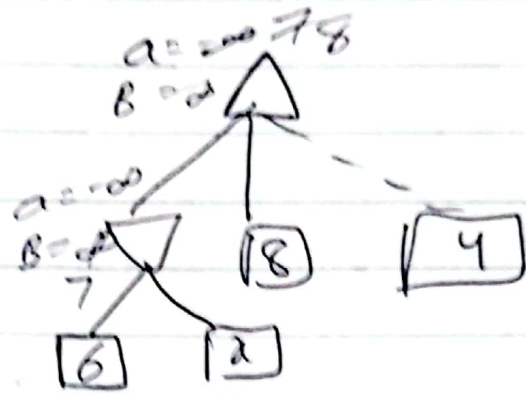
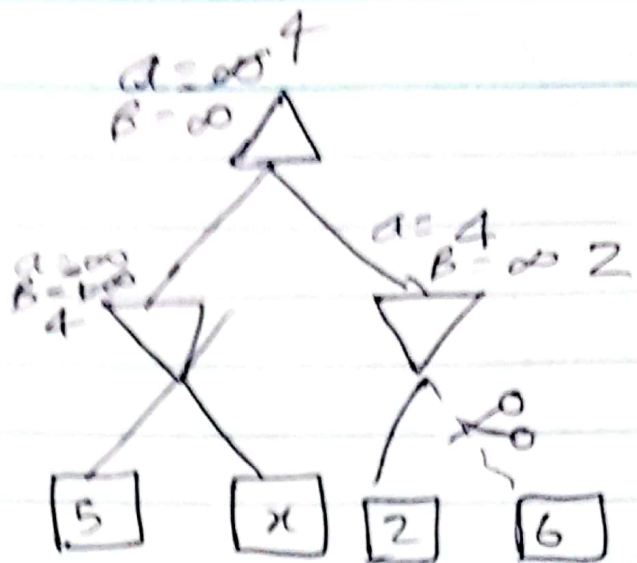
Update weights $[-2 \ -3 \ -3] - [1 \ 2 \ 3] = [-3 \ -5 \ -6]$

③ $-3 \times 1 - 5 \times 2 - 6 \times 1 = -19 = -1$ Correct

④ $-3 \times 1 - 5 \times 3 - 6 \times 2 = -30 = -1$ Correct

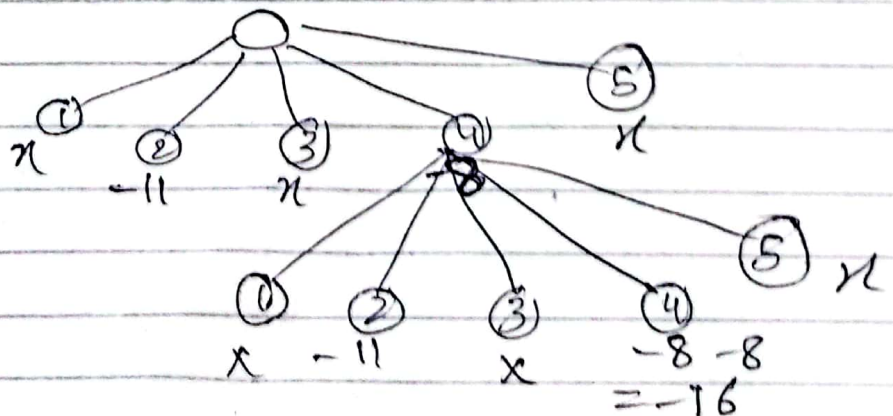
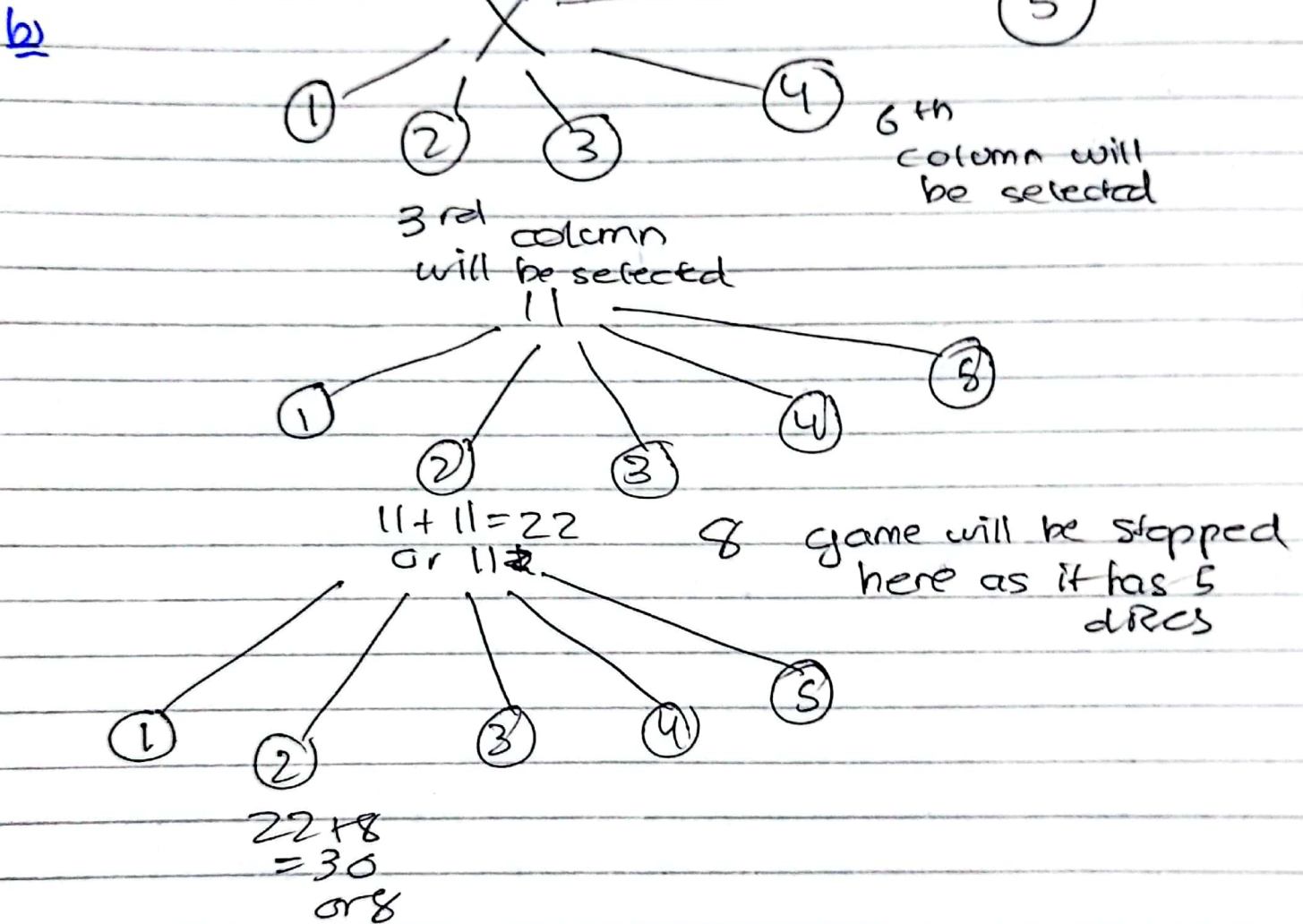
Question 5

1. None
2. 2 or greater than 2
3. 3 or greater than 3
4. None



Question 6

a) Nodes = $7^0 + 7^1 + 7^2 + \dots + 7^{42}$
 $= O(7^{42})$



Question 7

i)

	C1	C2	C3
A1 (2, 10)	0 ✓	3.6	8.06
A2 (2, 5)	5	4.2	3.16 ✓
A3 (8, 4)	8.4	5 ✓	7.28
A4 (5, 8)	3.6	0 ✓	7.21
A5 (7, 5)	7.07	3.6 ✓	6.7
A6 (6, 4)	7.21	4.12 ✓	5.38
A7 (4, 2)	8.06	7.21	0 ✓
A8 (4, 9)	2.23 ✓	1.41 ✓	7.61

ii)

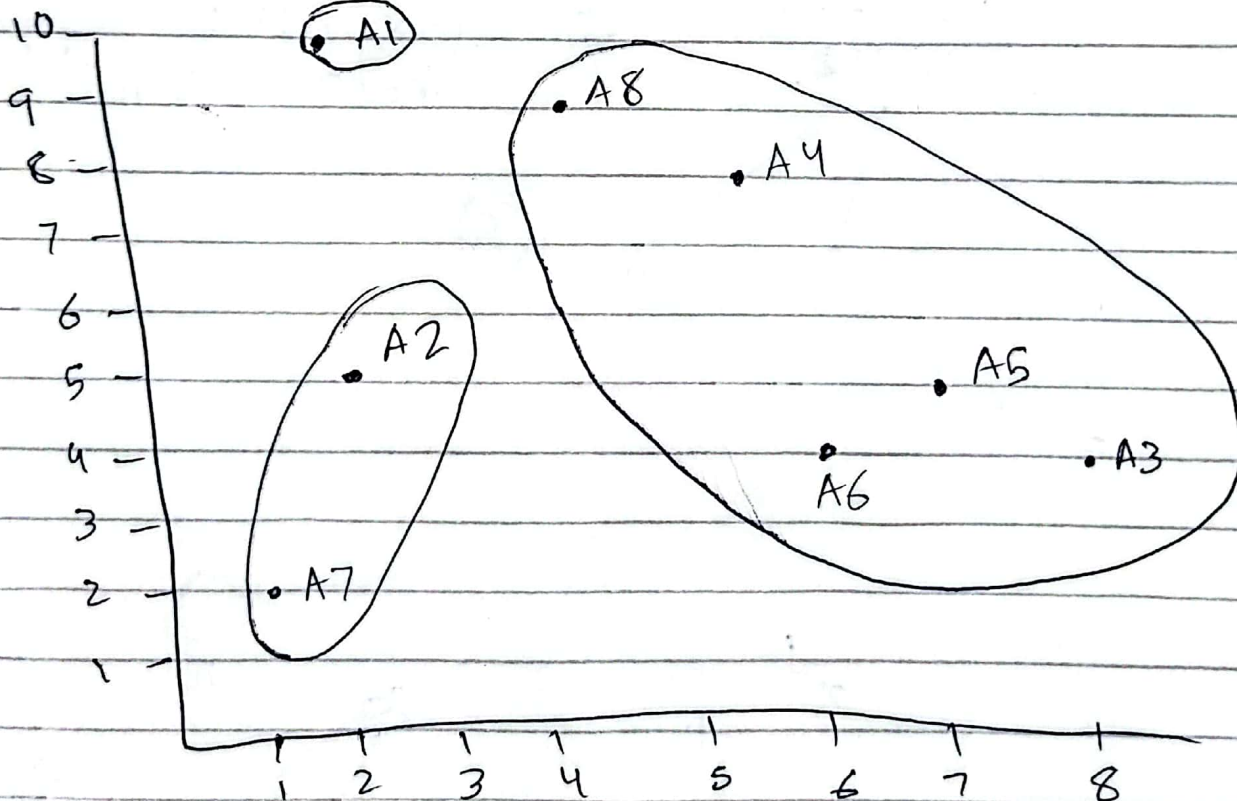
ii) Cluster 1 = A1 Cluster 2 = A3, A4, A5, A6, A8 Cluster 3 = A2, A7

Means for Cluster 1 = 2, 10

Means 11 11 2 = $\frac{8+6+7+6+4}{5}, \frac{4+8+5+9+9}{5} = 5.4, 6$

11 11 3 = $\frac{2+1}{2}, \frac{5+2}{2} = 1.5, 3.5$

iii)



Question 8

a) Inputs = 5

Output Types (Grades) = 2

Hidden layer = 2

(linear activation function)

Output = 1

(sigmoid activation function)

We can have 5 column train data at independent variables
And 1 column of train dependent data the data can
be like this

Proj-cont, Successor_Ratio_Average - Complete-Tim-Avg,
Skill_level, Overall-Exp - Grade

b) For Employee 1

$$= 5 \times -0.2 + -0.4 \times 0.5 + 0.8 \times 0.5 + -0.1 \times 0.5 + 0.3 \times 0.5$$
$$h_1 = -0.7$$

$$h_2 = 1 \times 1 + -0.4 \times 1 + 0.8 \times 1 + -0.1 \times 1 + 0.3 \times 1$$
$$h_2 = 0.6$$

Output layer

$$(-0.7)(2) + (0.6)(0.1) + 0.1 = 0.244$$

after applying sigmoid on answer

Grade = 0 = SATISFACTORY

For Employee 2

$$h_1 = 0.3$$

$$h_2 = 1$$

$$\text{Output} = 0.8 \Rightarrow 6(0.8) = 0.7 \quad \text{Grade} = 1$$

Correct

$$\textcircled{c)} \begin{pmatrix} -0.2 & 0.5 & 0.5 & 0.5 & 0.5 \\ 0 & 1 & 1 & 1 & 1 \end{pmatrix} \cdot \begin{pmatrix} 4 & -0.5 & 0.3 & -0.2 & 0.5 \end{pmatrix} = \begin{pmatrix} 4 & -0.5 & 0.3 & -0.2 & 0.5 \end{pmatrix}$$

$$\text{New weights} = \begin{pmatrix} -4.2 & 1 & 0.2 & 0.7 & 0 \\ -4 & 1.5 & 0.7 & 1.2 & 0.5 \end{pmatrix}$$

d) To handle three different grades, we can have 3 output neurons each representing 1 grade. Also, apply a softmax activation function at the output layer to produce a probability distribution over the 3 grades.

